INCH-POUND
MIL-DTL-915/66D
22 August 2002
SUPERSEDING
MIL-C-915/66C
29 September 1989

DETAIL SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, 59-CONDUCTOR, FOR OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE MSPW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-915 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

Conductor size 22-7 (16 required):

First - Copper conductor, in accordance with ASTM B 286, size 22-7, tin coated.
Second - Fluoropolymer insulation, nominal thickness 0.044 inch. (Taped insulation may be used.)
Third - Braided shield of AWG No. 34 or No. 36 copper. Minimum coverage 85 percent; braid angle 30 to 35 degrees.
Fourth - Shield insulation of one polyester tape, in accordance with NEMA FI 4, opaque white, with a jacket of transparent polyamide in accordance with ASTM D 4066. Nominal thickness 0.003.
- Alternate shield insulation of two polyester tapes, in accordance with NEMA FI 4, sealed. Inner tape opaque white; outer tape yellow.

For either type of shield insulation, use standard identification code applied by method 2, on the inner tape, using numbers 30 through 45.

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Conductor size 20-7 (18 required):

First - Copper conductor in accordance with ASTM B 286 size 20-7, tin coated.
Second - Polyvinyl chloride insulation (105°C), colored white. Nominal thickness 0.022 inch.
Third - Insulation jacket of clear polyamide in accordance with ASTM D 4066. Nominal thickness 0.003 inch.
Fourth - Braided shield of AWG No. 34 or 36 copper, tin coated, minimum coverage of 85 percent; braid angle of 30 to 35 degrees.
Fifth - Shield insulation consisting of one polyester tape, in accordance with NEMA FI 4, with a jacket of clear polyamide. Nominal thickness 0.003 inch.
- Alternate shield insulation of two polyester tapes, in accordance with NEMA FI 4, sealed.

For either type of shield insulation, use standard identification code applied by method 2, on the inner tape, using numbers 12 through 29.

Sixth - Conductors numbered 12, 13 and 14 formed into a triad. Nominal lay 3 inches.
Seventh - Conductors numbered 19 through 24 formed into three pairs, with the following pairing: numbers 19 and 20; numbers 21 and 22; numbers 23 and 24. Nominal lay 3 inches.

Conductor size AWG 16 (25 required):

First - Copper conductor, AWG size 16, class B stranding, tin coated.
Second - Polyvinyl chloride insulation (105°C). Nominal wall thickness 0.014 inch. Eleven conductors colored white; eleven conductors colored black; three conductors colored red.
Third - Jacket of clear polyamide in accordance with ASTM D 4066, over each conductor. Nominal thickness 0.003 inch.
Fourth - Sixteen conductors formed into eight pairs, each pair consisting of one white and one black conductor. Nominal lay 3 inches.
Fifth - Nine conductors formed into three triads, each triad consisting of one white, one red and one black conductor. Nominal lay 3-1/2 inches.
Sixth - Over each pair and each triad, a braided shield of AWG No. 34 or No. 36 copper, tin coated. Minimum coverage 85 percent; braid angle of 30 to 35 degrees.
Seventh - Shield insulation, over each shield, consisting of one polyester tape, in accordance with NEMA FI 4, with a jacket of clear polyamide. Nominal thickness 0.003 inch.

- Alternate shield insulation of two polyester tapes, in accordance with NEMA FI 4, sealed.

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For either type of insulation, use standard identification code applied by method 2, on the inner tape. The triads shall be numbered l, 2 and 3; the pairs shall be numbered 4 through 11.

Assembly:

First - A solid polychloroprene filler as the center of the core. Nonfibrous fillers shall be employed throughout the cable.
Second - Inner layer, consisting of the following:
(l) The shielded triads numbered l, 2 and 3.
(2) The triad consisting of the conductors numbered 12, 13 and 14.
(3) The conductors numbered 15, 16, 17 and 18.
Third - A binder tape.
Fourth - Middle layer, consisting of the following:
(1) The eight shielded pairs numbered 4 through ll.
(2) The three pairs consisting of conductors numbered 19 and 20, 21 and 22, 23 and 24.
Fifth - A binder tape.
Sixth - Outer layer; consisting of the following.
(l) The conductors numbered 25 through 45, cabled consecutively.
Seventh - A binder tape.
Eighth - Polychloroprene jacket (black) nominal thickness 0.110 inch. Cable overall diameter 1.565-inches minimum, 1.635-inches maximum. Cable surface marking required.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	
Size 22 conductors	17.71
Size 20 conductors	11.03
Size 16 conductors	4.44
Voltage withstand - volts, root mean square, minimum	
Size 22 conductors	
Conductor to conductor	5000
Conductor to shield	3000
Shield to shield	200
Size 20 and 16 conductors	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	200
Insulation resistance - megohms/1000 feet, minimum	
Size 22 conductors	
Conductor to conductor	5000
Conductor to shield	5000
Shield to shield	100

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Size 20 and 16 conductor:
- Conductor to conductor 150
- Conductor to shield 150
- Shield to shield 100

Group A:

Visual and dimensional ..... No failure
Watertightness - see MIL-DTL-915, Table XXII, for limits of water leakage ..... No failure
Characteristic impedance (size 22 conductors only) -at l MHz, ohms ..... 55 ± 5
Capacitance (size 22 conductors only) -at MHz; pF/ft, maximum 27

Group B:

Drip - at 75 ± l°C ..... Zero
Cold bending, cable - at 0 ± 2°C; over 18 inch maximum diameter mandrel ..... No damage
Physicals (unaged)
- Insulation (polyvinyl chloride only (PVC))
  - Tensile strength - lbf/in^2, minimum 1800
  - Elongation - percent, minimum 100
- Jacket (cable)
  - Tensile strength - lbf/in^2, minimum 1800
  - Elongation - percent, minimum 300
  - Set - inch, maximum 30/64

Group C:

Flammability - inches, maximum 2
Physicals (aged)
- Insulation (polyvinyl chloride only (PVC))
  - Air oven
    - Elongation - percent of unaged, minimum. 90

Cable jacket
- Air pressure heat
  - Tensile strength - percent of unaged, minimum 65
  - Elongation - percent of unaged, minimum 65
- Hot oil immersion
  - Tensile strength - percent of unaged, minimum 65
  - Elongation - percent of unaged, minimum. 65

Permanence of printing (jacket) - cycles, minimum 250
Cable filler removability ..... No failure
Shield-conformance to material coverage and construction ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cable aging and compatibility (95 ± 3°C) ..... No failure

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UNIT ORDERING LENGTH:
500 feet (nominal).

PART OR IDENTIFYING NUMBER (PIN): The PIN consists of the letter M, the general specification number along with the specification sheet, and a dash number representing type and size. Refer to MIL-DTL-915 for an example.

M	**915**	**/66-**	**X-X**
Prefix for Military Specification	Specification Number	Specification Sheet Number	Type and Size

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy - SH
(Project 6145-2317-011)

Review activities:
Army – AR, CR
Navy - CG, EC
DLA - CC

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